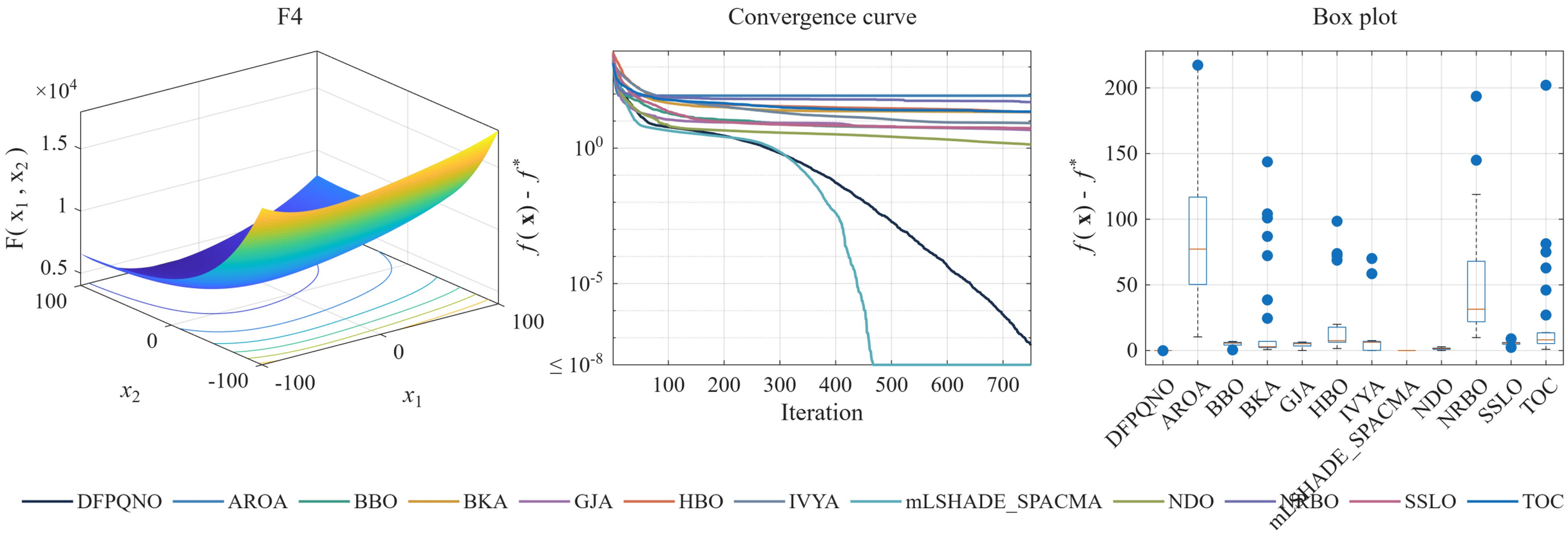

F4
×10^4
1.5
1
0.5
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^0
10^-5
≤ 10^-8
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
200
150
100
50
0
f(x) - f*
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC
DFPQNO
AROA
BBO
BKA
GJA
HBO
IVYA
mLSHADE_SPACMA
NDO
NRBO
SSLO
TOC